

Thomas Weng

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Robotics Research Scientist focusing on robot post-training for contact-rich manipulation:
finetuning pretrained policies for deployment scenarios through real-world interaction.

EDUCATION

Carnegie Mellon University

Ph.D. in Robotics

Advisor: Dave Held

Thesis Committee: Dave Held, Oliver Kroemer, Shubham Tulsiani, Alberto Rodriguez (MIT)

2018 - 2023

Pittsburgh, PA

Yale University

B.S. Computer Science & B.A. Economics

GPA: 3.77 / 4.0 with distinction in the C.S. major

Senior Thesis Advisor: Brian Scassellati

2011 - 2015

New Haven, CT

PUBLICATIONS

- [C12] Dodeja, L., Biza, O., Vats, S., Hart, S., Tellex, S., Walters, R., Schmeckpeper, K., and **Weng, T.** When Life Gives You BC, Make Q-functions: Extracting Q-values from Behavior Cloning for On-Robot Reinforcement Learning. *Robotics: Science and Systems (RSS)*, 2026.
- [C11] Schroeder, P., Biza, O., **Weng, T.**, Luo, H., Glass, J. ROVER: Recursive Reasoning Over Videos with Vision-Language Models for Embodied Tasks. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [C10] Dodeja, L., Schmeckpeper, K., Vats, S., **Weng, T.**, Jia, M., Konidaris, G., Tellex, S. Accelerating Residual Reinforcement Learning with Uncertainty Estimation. *IEEE Robotics and Automation Letters*, 2025. The contents of this paper were also selected by the ICRA 2026 Program Committee for presentation at the conference.
- [C9] Biza, O., **Weng, T.**, Sun, L., Schmeckpeper, K., Kelestemur, T., . . . Wong, L. L. On-Robot Reinforcement Learning with Goal-Contrastive Rewards. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [C8] **Weng, T.**, Held, D., Meier, F., and Mukadam, M. Neural Grasp Distance Fields for Robot Manipulation. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [C7] Tirumala, S.*, **Weng, T.***, Seita, D.*, Kroemer, O., Temel, Z., and Held, D. Learning to Singulate Layers of Cloth Based on Tactile Feedback. *IEEE International Conference on Intelligent Robots and Systems (IROS)*, 2022.
- [C6] **Weng, T.**, Bajracharya, S., Wang, Y., Agrawal, K. and Held, D. FabricFlowNet: Bimanual Cloth Manipulation with a Flow-based Policy. *Conference on Robot Learning (CoRL)*, 2021.
- [C5] Qian, J.*, **Weng, T.***, Okorn, B., Zhang, L., and Held, D. Cloth Region Segmentation for Robust Grasp Selection. *International Conference on Intelligent Robots and Systems (IROS)*, 2020.
- [J1] **Weng, T.**, Pallankize, A., Tang, Y., Kroemer, O., and Held, D. Multi-modal perception and transfer learning for grasping transparent and specular objects. *IEEE Robotics and Automation Letters*, 2020. The contents of this paper were also selected by the ICRA 2020 Program Committee for presentation at the conference.
- [C4] **Weng, T.**, Perlmutter, L., Nikolaidis, S. , Srinivasa, S., and Cakmak, M. Robot Object Referencing through Situated Legible Projections. *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.
- [C3] Sefidgar, Y.*, **Weng, T.***, and Cakmak, M. RobotIST: Interactive Situated Tangible Robot Programming. *Proceedings of the Symposium on Spatial User Interaction*. ACM, 2018.
- [C2] Admoni, H., **Weng, T.**, and Scassellati, B. Modeling communicative behaviors for object references in human-robot interaction. *IEEE International Conference on Robotics and Automation (ICRA)*, 2016.
- [C1] Admoni, H., **Weng, T.**, Hayes, B. and Scassellati, B. Robot nonverbal behavior improves task performance in difficult collaborations. *ACM/IEEE International Conference on Human Robot Interaction (HRI)*, 2016.

RESEARCH AND INDUSTRY EXPERIENCE

Robotics and AI Institute Robotics Research Scientist <i>Researching sample-efficient skill learning for manipulation [C9-12].</i>	2023 - present
Facebook Research and AI Mentorship Program Visiting Researcher with Dr. Mustafa Mukadam <i>Published on neural field representations for manipulation through a collaborative CMU-Meta program [C8].</i>	2021 -2023
University of Washington Human-Centered Robotics Lab Research Scientist with Prof. Maya Cakmak <i>Published papers on tangible robot programming and robot-mounted projectors for human-robot interaction [C3, C4].</i>	2017 - 2018
Microsoft Corp., AI and Research Software Engineer on Bing <i>Worked on Bing Answers for enterprise Q&A, flight booking, and the 2016 presidential election.</i>	2015 - 2017
Yale University Social Robotics Lab Undergraduate Researcher with Prof. Brian Scassellati <i>Published papers on modeling and generating robot non-verbal gestures [C1, C2].</i>	2014 - 2015

AWARDS AND HONORS

RSS Pioneer, Robotics: Science and Systems (RSS)	2023
Graduate Research Fellowship Award, National Science Foundation	2019
Graduate Research Fellowship Honorable Mention, National Science Foundation	2018
Computer Science Research Prize, Yale University	2015
Trumbull College Scholarship for Economics, Yale University	2014
Maher Family Scholarship, Yale University	2013, 2014

MENTORSHIP AND TEACHING

Lakshita Dodeja, RAI Intern	2025 - 2026
Philip Schroeder, RAI Intern	2025 - 2026
James Staley, RAI Intern	2025
Arpit Bahety, RAI Intern	2025
Mansi Agrawal, M.S. Robotics	2021 - 2023
Sashank Tirumala, M.S. Robotics	2021 - 2023
Sujay Bajracharya, M.S. Robotics	2020 - 2021
Rashmi Anil, CMU undergraduate	2019 - 2020
Khush Agrawal, RI Summer Scholar	Summer 2020
Yimin Tang, RI Summer Scholar	Summer 2019
Amith Pallankize, visiting undergraduate	2018 - 2019
Teaching Assistant, CMU 16-720 Computer Vision	Spring 2021
Teaching Assistant, CMU 16-811 Math Fundamentals for Robotics	Fall 2020
Mentor, CMU Masters in Research and Software Development (MRSD) Team	2018 - 2019
Teaching Assistant, University of Washington CSE 481C Robotics Capstone	Spring 2017

SERVICE AND LEADERSHIP EXPERIENCE

Reviewer

Robotics: Science and Systems (RSS)
Conference on Robot Learning (CoRL)
Robotics and Automation Letters (RA-L)
International Conference on Robotics and Automation (ICRA)
International Conference on Intelligent Robots and Systems (IROS)
Human Robot Interaction (HRI)
Humanoids
Transactions on Mechatronics
NeurIPS Workshop: Black in AI
Masters in Computer Vision (MSCV) Admissions, 2020

Graduate Student Assembly (GSA) Representative , Carnegie Mellon University <i>Elected representative of graduate students at the CMU Robotics Institute.</i>	2018 - 2023
Legislative Action Days Delegate <i>Met with senatorial and congressional staff in Washington, D.C. to advocate on behalf of graduate students on policy issues such as research funding, immigration reform, food insecurity, mental health, and more.</i>	2019, 2021
AI Undergraduate Research Mentor , CMU <i>Met monthly with an undergraduate from an underrepresented background on conducting research at CMU.</i>	2019
Girls of Steel FIRST Robotics Team Mentor , CMU Robotics Institute <i>Mentored the all-girls, multi-school team for the FIRST Robotics competition.</i>	2018
Tour Manager , Yale Alley Cats a cappella group <i>Managed tours for 14 undergraduates to the American South, the Pacific Northwest, Europe, and Asia. The Yale Alley Cats is one of the nation's most well-traveled a cappella groups.</i>	2012 - 2014

INVITED TALKS AND OUTREACH

Learning Representations for Manipulating Everyday Objects Robotics and AI Institute (formerly Boston Dynamics AI Institute)	September 2023
Visual Action Representations for Contact-Centric Manipulation University of Pennsylvania	September 2023
New York University	September 2023
Columbia University	August 2023
UC San Diego	August 2023
Google x Prep for Prep x VEX Robotics Workshop Guest Speaker , New York, NY <i>Presented on robotics to 100 gifted students of color in 6th-10th grade.</i>	2022, 2023
Buckley School Guest Speaker , New York, NY <i>Gave presentations on robotics and tech career paths to 7th-9th graders.</i>	2018, 2020
Code Haven at Yale Guest Speaker , New Haven, CT <i>Spoke with students at under-served New Haven public schools about STEM careers.</i>	2017

TECHNICAL SKILLS

Graduate Coursework

16-811	Mathematical Foundations for Robotics
16-720	Computer Vision
10-701	Machine Learning
16-711	Kinematics, Dynamics, and Control
10-703	Deep Reinforcement Learning and Control
16-782	Planning and Decision-making in Robotics
10-725	Convex Optimization
16-881	Deep Reinforcement Learning for Robotics (seminar)

Robotics

Languages	C/C++, Python, MATLAB
Machine Learning	PyTorch, Tensorflow
Tools	ROS, MoveIt!, OpenCV, MuJoCo, PyBullet, Nvidia FleX, Unity
Robots	Franka Panda, Sawyer, PR2, Fetch, Baxter, Kuka, Aldebaran Nao
Sensors	Azure Kinect DK, Kinect v2, Intel RealSense, Primesense

Web Development

Languages	JavaScript, C#.NET, Python, Ruby
Frameworks	React, Node.js, Django, Ruby on Rails